

Access, inequality, and the first stage

Three papers · one recent general-interest read, one EMS access note, and one methods paper. The following article pages are edited reading excerpts, not complete replacements for the originals.

LITERATURE PULSE

This issue's common thread is **access is not the same as equal benefit**. Broadband reaches rural provinces but initially widens income gaps where complementary skills and institutions are weak. Ambulances formally cover rural Connecticut, yet travel, stationing, and staffing leave response times roughly twice as long. And an instrument can shift treatment while identifying very different returns across people. Across all three, the average effect hides who actually receives the service, how intensely, and with what capacity to convert it into an outcome.

01 RECENT / GENERAL INTEREST · *The Annals of Regional Science* · 2025 · CC BY 4.0

Digital divide and income inequality: causal evidence from Italian provinces

Using all Italian provinces from 2014–2022, the authors apply a continuous-treatment difference-in-differences design to the rollout of 30 and 100 Mbps rural broadband. Expansion raises several measures of income inequality on average, with stronger effects in Southern and more marginalized provinces; northern areas appear more likely to benefit once coverage reaches a critical mass. The result is a useful warning that infrastructure access alone need not be equalizing when skills, firms, and complementary institutions are unevenly distributed. The causal interpretation depends on rollout intensity being conditionally independent of other province-level changes after the paper's controls and spatial adjustments.

02 EMS / ACCESS · *Research in Health Services & Regions* · 2024 · CC BY 4.0

Socioeconomic factors influencing rural–urban ambulance response-time disparities in Connecticut

Using town-level response times from Connecticut's 2016 EMS report, the study compares rural and non-rural places and tests whether median household income explains the gap. Rural mean response time is almost 13 minutes—roughly twice the non-rural mean—yet income is not significantly correlated with response time within either group. This points toward geography, stationing, staffing, and volunteer-service capacity rather than town income alone. Treat the result as descriptive: the data are old, town-level, and cross-sectional, so they cannot isolate causal mechanisms or within-town inequality.

03 METHODS · *Federal Reserve FEDS* · 2022 revision · U.S. public domain

ivcrc: An instrumental-variables estimator for the correlated random coefficients model

This paper and Stata module address instrumental-variables models in which causal effects differ across people and treatment choices are correlated with those effects. A conditional-rank control function and local linear regressions recover average partial effects that standard 2SLS may not. In the returns-to-schooling application, conventional IV implies about a 13% wage return per year of schooling, while the correlated-random-coefficients estimate is nearer 8%. The method is practical and transparent, but relies on a continuously distributed endogenous treatment, valid instruments, and control-function assumptions that should be defended substantively.

Editorial note: references and appendices were omitted where possible; no included article contributes more than 15 text pages. Full citations and source links remain in the catalog. AI-written synopses were checked against the included source pages.



Digital divide and income inequality: causal evidence from Italian provinces

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Abstract

The digital economy can function either as a catalyst to stimulate economic growth or else as a driver of socioeconomic inequality when its benefits are unevenly distributed. This study investigates the effect of rural digital connectivity on income inequality in Italy. Utilizing NUTS 3 panel data spanning 2014–2022, we conduct a counterfactual Difference-in-Differences approach with continuous treatment intensity to estimate the impact of introducing rural broadband coverage at speeds of 30 and 100 Mbps on multiple measures of income distribution, including the Gini, Theil, and Atkinson indices. The empirical framework incorporates a comprehensive set of socioeconomic controls, as well as provincial and time fixed effects, to account for unobserved heterogeneity and regional path dependencies. Our findings indicate that broadband expansion is significantly associated with increasing inequality, suggesting that access alone does not guarantee inclusive outcomes, particularly in localities characterized by structural fragility and limited human capital. Additional heterogeneity and spatial analyses demonstrate that these inequality effects are more evident in southern provinces and localities with a higher concentration of inner areas, where the digital divide remains more pronounced. These findings accentuate the dual role of digitalization and highlight the necessity of coordinated policy interventions that combine infrastructure investment with digital skills development, institutional capacity-building, and spatially integrated governance strategies.

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1 Introduction

The rapid development of digital technologies over the past two decades has profoundly transformed the global economic landscape, reshaping labor markets, education systems, and public service delivery (Card et al. 2018; Acemoglu and Restrepo 2022; Montresor and Vezzani 2023; Capello et al. 2024; Acemoglu 2025). As digital tools increasingly underpin economic, social, and civic participation, access to digital sources has evolved from being a purely technical issue into a pivotal determinant of opportunity and inclusion (Lythreath et al. 2022).

When digital resources are unevenly distributed and access is not universally guaranteed, this phenomenon is conceptualized as the “digital divide,” illustrating the gap between those who have reliable access to information and communication technologies (ICT) and those with limited or no access (Soomro et al. 2020; Bergantino et al. 2025). Diverse forms of digital inequality can exacerbate broader social disparities by either inhibiting or enhancing individuals’ social and economic capital and their ability to engage meaningfully in social life (Wu et al. 2024; Ragnedda 2017). Given the extensive applications of digital technologies and the economic activity they enable, access to these tools has emerged as a pivotal determinant of territorial wealth and development.

Communities experiencing digital poverty—characterized by insufficient access to digital infrastructure, devices, or skills—are increasingly susceptible to many types of economic deprivation (Lagakos 2020; Barbero and Rodríguez-Crespo 2022; Bergantino et al. 2025). Inequality of access to digital resources can reinforce existing regional disparities and generate new forms of intra-regional inequality (Setthasuravich and Kato 2020).

The relevant literature attests to this dual potential of digital technologies. On the one hand, they can serve as a digital dividend, promoting inclusion and development within specific territories. Conversely, if access and digital skills are concentrated among already advantaged groups or localities, digitalization may exacerbate pre-existing inequalities, functioning as a digital divide (Peng and Dan 2023). Insufficient access to digital infrastructures and services can, therefore, translate directly into income inequality, as disadvantaged areas face limited opportunities for productivity growth, employment, and social mobility (Bourguignon 2022; Consoli et al. 2023; Feng et al. 2025).

Italy presents an exemplary context for analyzing the territorial and economic consequences of digital inequality. The country manifests a striking structural divide between Northern and Southern regions (Brida et al. 2014; Felice 2019; Monturano et al. 2025a, b), as well as between urban and rural areas. These disparities are demonstrated by patterns of digital access and usage.

In relation to connectivity, Italian territories are classified into three categories—black, gray, and white areas—based on the number of private operators providing broadband services (Presidenza del Consiglio dei Ministri 2015, 2021). “White areas” signify territories where no private operator is currently providing, or

planning to provide within the next three years, broadband services with speeds of at least 30 Mbps. Under national and EU regulations, these “market failure zones” are eligible for public interventions through dedicated investment programs (Law No. 55/2019, National Ultra-Broadband Plan). In contrast, gray areas are served by a single operator, while black areas host at least two competing operators.

The concentration of “white areas” in rural and internal territories demonstrates how inadequate digital infrastructure can exacerbate existing socioeconomic disparities, exposing these regions to digital poverty and marginalization. In Southern Italy, only 75.6% of households have home Internet access, compared to 86.6% in the Center and 89.8% in the North. Foundational digital competencies are similarly uneven, with only 32.3% of residents in the South possessing basic digital skills, versus 48.3% in the Center and 54.7% in the North (ISTAT 2023). These gaps in connectivity and digital ability reflect broader inequalities in infrastructure, education, and income (Reljic et al. 2021; ISTAT 2023).

The persistent territorial imbalances in digital access and skills underscore critical questions in relation to the distributive consequences of digitalization. This study investigates the effects of ultra-fast broadband expansion on income inequality across Italian provinces, with particular attention to heterogeneous and spatially interdependent impacts.

The analysis aims at achieving three principal objectives. First, it estimates the causal effect of broadband diffusion on income inequality, exploiting cross-territorial variation in digital coverage. Second, it examines whether the distributive outcomes of broadband expansion are mediated by local contextual factors. Third, it evaluates spatial spillovers, assessing whether broadband-induced effects propagate to neighboring provinces.

To achieve these goals, the study adopts a Difference-in-Differences (DiD) framework with continuous treatment intensity (Callaway et al. 2024), leveraging the quasi-experimental setting provided by the 2019 National Ultra-Broadband Plan (Law No. 55/2019). This policy targeted “white areas” -rural and peripheral municipalities excluded from private investment-thereby generating exogenous variation in broadband deployment intensity. The empirical analysis spans 2014–2022 and employs provincial (NUTS 3) data enriched with socioeconomic, demographic, and institutional indicators capturing local absorptive capacity for digital infrastructure. The research design links broadband diffusion to inequality via productivity, structural, and redistributive mechanisms as discussed in Sect. 3.

This study contributes to the literature by providing a causal assessment of broadband expansion and income inequality at the provincial level in Italy, integrating digital economics and regional development perspectives. It offers policy-relevant evidences on how local conditions shape the inclusiveness of digital transformation, informing debates on the EU Digital Decade and Italy’s national digital transition strategies, and clarifying whether digitalization functions as a convergence-enhancing force or as an amplifier of regional disparities.

The remainder of the paper is organized as follows. Section 2 reviews the literature on digitalization and inequality; Sect. 4 describes the data and main variables; Sect. 5 outlines the empirical strategy; Sect. 6 presents the results; and Sect. 7 concludes.

3 Theoretical framework: mechanisms linking digitalization and income inequality

Digitalization represents one of the most profound structural transformations of the contemporary economy. Its effects extend well beyond aggregate economic growth, reshaping the spatial and functional distribution of opportunities across regions and sectors (Bresnahan and Trajtenberg 1995; Bresnahan et al. 2002; Acemoglu and Restrepo 2018, 2022). Broadband diffusion acts as an enabling technology that reduces communication costs, enhances productivity, and expands access to markets and services (González et al. 2016; Mack et al. 2024; Briglauer et al. 2025). Yet, its distributive effects remain highly heterogeneous (Antonietti et al. 2025; Briglauer et al. 2025). Digital infrastructures may foster convergence by diffusing knowledge and integrating peripheral areas, or alternatively, may reinforce divergence by amplifying the cumulative advantages of already developed regions (Tranos 2012; Barbero and Rodríguez-Crespo 2022). Following the classical insights of Myrdal and Sitohang (1957) and Krugman (1991), technological progress tends to cluster spatially, generating cumulative causation processes that sustain economic polarization. Thus, digitalization may constitute a new driver of income and spatial inequality (Malecki 2002; Tranos et al. 2021).

The literature identifies three main mechanisms through which digitalization affects income inequality: (1) productivity differentials (Bourguignon 2022; Dedola et al. 2023), (2) structural transformation (Matthess and Kunkel 2020), and (3) redistribution of economic gains (Richmond and Triplett 2018; Nayyar et al. 2024; Xiao et al. 2024).

First, broadband expansion tends to augment overall productivity, yet the distribution of these gains is uneven across space. Advanced regions, characterized by innovative firms, complementary infrastructure, and high levels of human capital, are better positioned to fully exploit digital opportunities (Resce et al. 2022; Peng and Dan 2023; Zinilli and Cerulli 2023). Conversely, less developed areas, which lack these assets, frequently adopt digital technologies passively and capture only limited benefits. Consequently, digitalization generates asymmetric returns that exacerbate income gaps both between and within territories. The digital divide is thus not only a matter of mere access but encompasses the local capacity to convert digital inputs into inclusive growth, with significant implications for income distribution and welfare (Tranos et al. 2021; Antonietti et al. 2025).

Second, digital technologies reshape the economic structure by promoting high-value-added, knowledge-intensive sectors that are typically concentrated in metropolitan and more advanced areas (Florida 2005; Brynjolfsson and McAfee 2014). The spatial clustering of digital services, data-driven industries, and creative activities generates new agglomeration dynamics, reinforcing divergence between core and peripheral territories (Carlsson 2004; Tranos et al. 2021). The digital transition therefore produces a spatially selective structural shift: regions capable of specializing in advanced sectors capture an increasing share of value added, while others remain locked in low-productivity trajectories. This selective transformation

4 Data and descriptive statistics

4.1 The dataset

To assess the impact of the introduction of broadband on income inequality in Italy, as reported in Table 1, we build a panel data set that includes all Italian provinces for the period 2014–2022. The main dependent variable is the Gini index, calculated from the income data of the Italian Ministry of Economy and Finance (MEF).

On the mathematical side, following Antulov-Fantulin et al. (2021), and Maranzano et al. (2025), we calculated the provincial level using aggregated tax data by income brackets and it is formally defined as:

Table 1 List of variables and corresponding data sources

Variables	Source
<i>Outcome variables</i>	
Gini index	Italian ministry of economy and finance (MEF)
Theil index	Italian ministry of economy and finance (MEF)
Atkinson index	Italian ministry of economy and finance (MEF)
Per capita income	Italian ministry of economy and finance (MEF)
<i>Independent variables</i>	
FTTH rural surface	AGCOM broadband coverage dashboard
FTTH rural population	AGCOM broadband coverage dashboard
Target 100 Mbps rural surface	AGCOM broadband coverage dashboard
Target 100 Mbps rural population	AGCOM broadband coverage dashboard
<i>Control variables</i>	
– <i>Socio-economic</i>	
Employment rate	ISTAT—employment by sector
Economic specialization	ISTAT—experimental statistics
Agricultural value added	ISTAT—national accounts by territory
Services value added	ISTAT—national accounts by territory
Cohesion fund	Opencoesione.it—Department for Cohesion Policies and the South
– <i>Demographic, territorial</i>	
Total population	ISTAT—demographic statistics
Foreign population	ISTAT—demographic statistics
Surface	ISTAT—geographic statistics
Tourism facilities (total)	ISTAT—tourism statistics
Tourism beds (total)	ISTAT—tourism statistics
– <i>Institutional and political</i>	
Pillar 1 Bureaucracy	Cerqua et al. (2025)
Pillar 2 Politics	Cerqua et al. (2025)
Pillar 3 Economy	Cerqua et al. (2025)

untreated ones), indicating a potential link between broadband expansion and economic performance.

With regard to control variables, treated provinces tend to exhibit higher employment rates and the most relevant value added in both services and agriculture. This is consistent with the fact that broadband infrastructure is more likely to be deployed in economically dynamic territories. However, untreated provinces are characterized by larger average population sizes, both total and foreign, highlighting that the population scale does not necessarily drive the treatment target.

Institutional and political indicators (Tables 1–3) show only limited variation between the two groups, suggesting that the institutional framework is relatively homogeneous between provinces and therefore less likely to confound the effect of broadband coverage.

Table 3 provides an aggregated breakdown at the regional level of the Gini index and broadband connectivity measures. Moreover, it highlights significant differences even when comparing regions, both in terms of economic inequality and digital coverage. The data reveal considerable regional heterogeneity.

Southern regions such as Calabria (0.22) and Sicily (0.26) show the lowest levels of inequality, whereas northern and central regions such as Trentino-Alto Adige (0.37) and Valle d'Aosta (0.35) display the highest values of inequality. When considering macro-areas, the North reports a higher average Gini index (0.32) compared to the South (0.27). This regional gap is even wider than the difference observed between the internal and the centers, where the disparity is only 0.03.

With regard to digital indicators, the average rural FTTH coverage exhibits wide disparities across regions. For example, Basilicata and Molise show relatively high levels of rural FTTH diffusion, whereas Liguria, Sardinia, and Valle d'Aosta are at the bottom of the distribution. The 'Target 100 Mbps' indicator follows a similar trend. The national average is still quite low, indicating that ultra-fast broadband deployment in rural areas is still at an early stage. However, some regions such as Lazio and Emilia-Romagna stand out for their relatively high levels of coverage, while several Southern and Island regions lag behind. Looking at the aggregated values at the bottom of the table, these confirm the existence of such disparities. The South shows a higher average rural broadband coverage per region (driven by specific cases such as Basilicata and Puglia).

In summary, this table highlights the coexistence of territorial inequalities in income distribution and pronounced digital divides in broadband diffusion, with substantial heterogeneity not only between the North and South, but also within each macro-area. These descriptive patterns provide initial evidence of the potential uneven effects of broadband expansion in rural areas of Italy.

5 Empirical strategy

To assess the impact of ultra-fast broadband diffusion on income inequality at the provincial level, we employ an econometric identification strategy, based on a Difference-in-Differences design with continuous treatment (Callaway et al. 2024).

This empirical strategy is directly informed by the theoretical framework outlined in Sect. 3, which identifies three mechanisms—increased productivity, structural transformation, and the redistribution of economic gains—through which broadband diffusion may influence income distribution. Specifically, our analysis examines whether the diffusion of ultrafast broadband has mitigated or exacerbated income inequality across Italian provinces, while accounting for the structural and institutional conditions emphasized in the conceptual model.

The empirical design compares changes in inequality indicators before and after the launch of the national fiber coverage programs, exploiting variation in coverage intensity across Italian provinces. We define 2014–2018 as the pre-treatment period and 2019–2022 as the post-treatment period, corresponding to the rollout and consolidation of the national ultra-fast broadband extension program, with particular focus on rural areas. In 2019, public policies targeting the so-called white areas—rural and peripheral municipalities that had previously been largely excluded from fiber infrastructure—became operational (Presidenza del Consiglio dei Ministri 2015, 2021). From this point onward, network expansion no longer followed market-driven dynamics but was directed by a centrally coordinated national strategy. Accordingly, the exogeneity assumption is grounded in the institutional design of this program. The national ultra-broadband plan was implemented by means of centrally managed tenders and public–private partnerships, in which the allocation of public funds and the sequencing of deployment were determined by administrative and technical criteria, such as existing infrastructure, geographical constraints, and project feasibility, rather than by local economic performance or inequality trends. Consequently, all provinces were subject to the policy simultaneously, yet they differed in terms of both the intensity and timing of broadband diffusion. This institutional arrangement creates a quasi-experimental setting in which variation in coverage intensity can be interpreted as an exogenous source of treatment exposure, thereby enabling the identification of the causal effect of broadband expansion on income inequality. Within this framework, our approach follows a *dose–response function* perspective (Cerulli and Ventura 2021), where the continuous treatment variable quantifies the degree of fiber optic diffusion in each province, weighted by population and surface area.

The credibility of our causal estimates relies on the standard identifying assumptions of the DiD framework (Marcus and Sant’Anna 2021; Rambachan and Roth 2023; Roth and Sant’Anna 2023). Since all provinces were exposed to the policy after 2019, identification is achieved through differences in the intensity and timing of broadband deployment across territories rather than via a binary comparison of treated and untreated units. In this continuous-treatment setting, the key assumption is that, in the absence of the policy intervention, provinces would have exhibited parallel trends in income inequality. To assess the plausibility of this assumption, we conducted a pseudo-parallel-trends analysis based on the evolution of the outcome variable. Provinces were grouped into binary (above and below the median) and tertile classes according to their ex post inequality levels, as measured by the Gini index over the post-treatment period (2019–2022). We then compared the pre-treatment trajectories of the average Gini coefficient across these groups. The resulting trajectories indicate highly similar pre-2019 dynamics, supporting the plausibility of

As explained in the Data Section (4), in our analysis the main outcome variables is the Gini index, which constitutes our principal measure of inequality, reflecting disparities in the income distribution within each province (Liberati and Resce 2022; Maranzano et al. 2025). To ensure robustness, we re-estimate the main specifications using alternative measures of inequality, namely the Theil index, the Atkinson index, and the per capita income level (De Clercq et al. 2023). These alternative indicators allow us to verify that the results do not depend on the specific inequality measure chosen.

To explore potential territorial heterogeneities, we further stratify the analysis along several dimensions. First, we distinguish between provinces above and below the median share of municipalities classified as internal areas under the SNAI strategy,² and between provinces located in the Center-North and those in the South. Second, we divide the sample into terciles of the Gini index to assess whether the effects of digital diffusion differ across low, medium, and high inequality contexts. This stratified design allows for a more nuanced understanding of how digitalization interacts with existing territorial disparities.

To account for effects of potential territorial spillovers, we extend the benchmark DiD specification within a spatial econometric framework following the established literature (see for example, LeSage and Pace 2009; Arbia 2014; Elhorst 2014; Anselin 2022) (Chagas et al. 2016; Kosfeld et al. 2021). The panel structure of our dataset—combining the cross-sectional dimension (provinces) with the temporal dimension (years)—permits us to estimate spatial panel models with fixed effects where dependence can occur either through the dependent variable, as in the Spatial Autoregressive Model (SAR), or through the error term, as in the Spatial Error Model (SEM). The Spatial Durbin Model (SDM) is employed as the most general specification, allowing both forms of dependence to be incorporated simultaneously and enabling the joint estimation of direct and spillover effects among provinces.

This spatial extension is directly grounded in the theoretical framework, which posits that broadband-induced economic gains or losses can propagate across neighboring territories through cumulative and feedback effects. By capturing these spillover dynamics, the spatial model provides a more comprehensive empirical representation of the causal mechanisms underlying regional inequality. In particular, it allows us to test whether digital infrastructure investment in one province generates measurable distributive effects in contiguous areas, thereby elucidating the systemic and interdependent nature of regional development processes.

In general, the spatial panel specification can be written as:

$$\begin{aligned} \text{Ineq}_{it} &= \theta_i + \lambda_t + \rho W\text{Ineq}_{it} + \text{Digital}_{it}\beta + W\text{Digital}_{it}\theta + X_{it}\gamma + u_{it}, \\ u_{it} &= \lambda W u_{it} + \varepsilon_{it}, \end{aligned} \quad (7)$$

² The SNAI classification is a national law that introduced a road metric that divides municipalities into inner areas and centers, based on accessibility to essential services.

where: Ineq_{it} represents the Gini index in province i in year t ; θ_i ; and λ_t are provincial and temporal fixed effects, respectively; Digital_{it} is the ultra-broadband diffusion indicator; X_{it} is the vector of socio-economic covariates, and W is the spatial adjacency matrix. The parameters ρ , θ , and λ capture the spatial dependence of the dependent variable, that of the explanatory variables, and the autocorrelation of the errors, respectively.

For the economic interpretation of the estimated coefficients, following LeSage and Pace (2009), and Elhorst (2014), we decompose the effects using the spatial multiplier matrix. The marginal effect of a change in the digital variable in province i is given by:

$$\frac{\partial \text{Ineq}}{\partial \text{Digital}_k} = (I - \rho W)^{-1} (\beta_k I + \theta_k W), \quad (8)$$

where the diagonal represents the direct effects (average impact of local ultra-broadband diffusion on provincial inequality), while the extra-diagonal elements measure the indirect or spillover effects (average impact of diffusion in neighboring provinces on local inequality). The sum of the two components gives the total effect. Furthermore, we calculate direct, indirect, and total effects at the province level, both globally and in the post-treatment period (2019–2022), and also provide impact decompositions conditional on the full set of covariates, the latter of which are presented in the Supplementary Materials to this paper. This allows us to assess whether the spatial impacts of broadband policies are robust over time and across different model specifications. This empirical design is consistent with the spatial econometrics literature on public policies and territorial outcomes (e.g., Arbia 2006; Elhorst 2014; Annoni et al. 2019; Panzera and Postiglione 2022,) and allows us to distinguish the effect of ultra-broadband diffusion at the provincial level from that transmitted through territorial interdependence mechanisms, providing a more complete and consistent picture of the link between digitalization and inequality.

6 Results

In this section, we discuss the main econometric results of the DiD estimation, the robustness analyzes, and the spatial estimates for studying spillovers.

6.1 Main results

Table 4 reports the empirical results of the main analysis examining the effect of ultra-broadband expansion-measured through two sets of treatment indicators, FTTH, and Target 100 Mbps in rural areas-on income inequality, measured by the Gini index (Models 1–4). Specifically, the Table 4 presents the benchmark DiD estimates across all model specifications (Spec. 1–4).

The results consistently attest to a positive and statistically significant effect across all specifications of Eq. 6. This finding suggests that the expansion of digital infrastructure in rural areas correlates with an increase in income inequality.

Table 4 DiD estimates on the Gini index

	Spec_1	Spec_2	Spec_3	Spec_4
<i>Model_1</i>				
FTTH_rural_surface	0.001 (0.000)	0.001 (0.000)	0.001* (0.000)	0.001** (0.000)
<i>Model_2</i>				
FTTH_rural_population	0.231*** (0.043)	0.207*** (0.043)	0.209*** (0.043)	0.2*** (0.041)
<i>Model_3</i>				
Target100_rural_surface	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)
<i>Model_4</i>				
Target100_rural_population	0.253*** (0.026)	0.232*** (0.027)	0.238*** (0.027)	0.224*** (0.026)
Socio-economic		YES	YES	YES
Demographic, territorial			YES	YES
Institutional and politics				YES
Time fixed-effects	YES	YES	YES	YES
Provincial fixed-effects	YES	YES	YES	YES
N. obs	951	913	913	913

All continuous predictors are mean-centered and scaled by 1 standard deviation. Significance levels:

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$

In Model 1, Specification 1 initially yields a smaller and statistically non-significant coefficient; however, the coefficient becomes statistically significant once the socio-economic (Specification 3) and institutional controls (Specification 4) are incorporated. This pattern highlights the importance of accounting for local characteristics as probable confounding factors when isolating the policy's causal impact.

Model 2 shows that a higher share of FTTH weighted by population, results in an increase in the Gini index, with a confidence interval of 99% across all four specifications. Models 3 and 4, which use the Target 100 Mbps indicators, show a similar overall pattern to Models 1 and 2.

Additionally, the stepwise inclusion of controls in Specifications 2–4 does not substantially change either the magnitude or significance of the estimated coefficients. This stability improves the robustness of the findings and suggests that the observed increase in inequality is unlikely to stem from pre-existing structural differences between treated and untreated areas. Rather, it appears to reflect a direct effect of broadband expansion. These findings align with previous research which shows that digital infrastructure development does not inherently reduce inequality. Indeed, in the absence of equitable access and adequate digital skills, such technological progress may exacerbate existing socioeconomic disparities (Tranos et al. 2021; Peng and Dan 2023; Antonietti et al. 2025).

Table 5 shows the results a forward-shifted lead analysis of the Gini index, where the dependent variable is shifted forward to assess whether the impact of broadband

Table 5 DiD estimates with the forward-shifted lead Gini index

	Spec_1	Spec_2	Spec_3	Spec_4
<i>Model_1</i>				
FTTH_rural_surface	0.001** (0.000)	0.001* (0.000)	0.001** (0.000)	0.001*** (0.000)
<i>Model_2</i>				
FTTH_rural_population	0.248*** (0.038)	0.219*** (0.038)	0.22*** (0.038)	0.211*** (0.037)
<i>Model_3</i>				
Target100_rural_surface	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)
<i>Model_4</i>				
Target100_rural_population	0.269*** (0.025)	0.244*** (0.025)	0.249*** (0.025)	0.238*** (0.025)
Socio-economic		YES	YES	YES
Demographic, territorial			YES	YES
Institutional and politics				YES
Time fixed-effects	YES	YES	YES	YES
Provincial fixed-effects	YES	YES	YES	YES
N. obs	1058	1015	1015	1015

All continuous predictors are mean-centered and scaled by 1 standard deviation. Significance levels: *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$

deployment persists over time rather than being confined to the short term. By estimating the effect of current broadband expansion on future levels of income inequality, this specification makes it possible to test whether the observed increase in disparities reflects a temporary adjustment or a more enduring structural shift.

The results corroborate the benchmark findings reported in Table 4. Across all specifications, the estimated coefficients remain positive, statistically significant, and comparable in magnitude. This consistency suggests that the inequality-enhancing effect of broadband expansion is not merely a transient phenomenon associated with the initial rollout phase, but endures into the medium term. The enduring nature of the effect is consistent with the operation of a cumulative mechanism: early adopters—typically higher-income households and firms—continue to reap disproportionate benefits from improved digital infrastructure, thereby consolidating their relative advantage, while less advantaged groups fail to close the gap at a similar pace.

Overall, this evidence supports the interpretation that, in the absence of complementary policies such as digital skills training, targeted subsidies, or redistributive interventions, broadband infrastructure may inadvertently reinforce pre-existing socioeconomic divides. In other words, digital access alone does not function as an equalizing force; without inclusive measures, it risks entrenching and perpetuating new forms of inequality that persist over time (Di Caro and Fratesi 2022; Steinebach 2022).

Furthermore, these results are consistent with the first mechanism outlined in the theoretical framework. Broadband expansion increases overall productivity but disproportionately benefits higher-income households and firms with stronger digital capabilities. Consequently, the diffusion of ultra-fast broadband reinforces unequal returns to technology adoption across territories.

6.2 Robustness check and heterogeneity analysis

To assess the robustness of the estimates recorded in Table 4, we conduct additional analyses utilizing alternative measures of income inequality. The results summarized in Table 6 are consistent with those obtained from the benchmark DiD analysis, based on the Gini index. Indeed, across all models and specifications, the alternative inequality indicators—specifically the Theil and Atkinson indices—yield positive and statistically significant coefficients. These results corroborate the robustness of the main findings and lend further support to the conclusion that the expansion of ultra-broadband infrastructure in rural areas significantly contributes to increasing inequality.

Similarly, the results for per capita income show a positive and statistically significant association, implying that although overall income levels may increase,

Table 6 DiD estimates with alternative inequality indicators (Theil, Atkinson, Per capita income)

	Theil	Atkinson	Income
<i>Model_1</i>			
FTTH_rural_surface	0.001*** (0)	0.001*** (0)	0.002*** (0)
<i>Model_2</i>			
FTTH_rural_population	0.249*** 0.03	0.131*** 0.013	0.258*** 0.037
<i>Model_3</i>			
Target100_rural_surface	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)
<i>Model_4</i>			
Target100_rural_population	0.227*** (0.019)	0.227*** (0.019)	0.2*** (0.024)
Socio-economic	YES	YES	YES
Demographic, territorial	YES	YES	YES
Institutional and politics	YES	YES	YES
Time fixed-effects	YES	YES	YES
Provincial fixed-effects	YES	YES	YES
N. obs	913	913	913

All continuous predictors are mean-centered and scaled by 1 standard deviation. Significance levels: *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$. Income represent the levels of per capita income for each municipalities

such growth is not evenly distributed across the population (Li et al. 2022; Wu et al. 2024).

Overall, these robustness analyses strengthen the credibility of our empirical evidence, showing that the observed effects are not merely artifacts of a specific inequality metric but instead reflect a broader and enduring pattern of widening disparities associated with digital infrastructure development in rural areas.

Taken together, these findings strengthen the interpretation that broadband-induced inequality stems from underlying structural transformation processes. The expansion of digital infrastructure tends to stimulate growth in higher value-added and service-oriented sectors-typically concentrated in more economically developed areas-thereby amplifying spatial polarization and reinforcing existing regional inequalities.

While the main results consistently show that broadband expansion in rural areas is directly associated with increasing income inequality, it is crucial to explore whether this relationship varies across different territorial and socioeconomic contexts. To this end, we conduct a series of heterogeneity analyses aimed at discovering how the impact of broadband diffusion differs across spatial scales. Specifically, we investigate: (1) whether the impacts structurally between more agglomerated provincial areas-characterized by higher service intensity-and more remote territories (inner areas vs centers); (2) whether regional disparities exist between macro-regions (North vs. South), reflecting Italy's well-documented economic dualism; and (3) whether the effects vary across different levels of pre-existing inequality as measured by terciles of the Gini index. The corresponding results are presented in Tables 7, 8 and 9.

Table 7 reports the results of the heterogeneity analysis for inner areas and centers. The results reveal substantial territorial differences in the impact of broadband expansion on inequality, depending on the territorial context. In the inner provinces (identified under the SNAI in Table 7), the coefficients associated with the FTTH coverage (both population and area weighted) are positive and statistically significant across all Models (1–4) and specifications. This evidence suggests that the introduction of broadband in more peripheral provinces tends to exacerbate income inequality, consistent with the notion that rural digitalization processes may amplify existing structural disadvantages (Philip and Williams 2019; Wang and Liu 2025).

Conversely, in the more agglomerated provinces (defined by the median share of inner municipalities), the results display a different pattern (see Table 7). The FTTH rural coefficients are consistently negative and statistically significant across all specifications (1–4). This suggests that in less marginalized provinces, broadband expansion is associated with reduced inequality. By contrast, the coefficients for the Target 100 Mbps indicators are positive but not statistically significant, implying that while ultra-fast broadband may be associated with increased inequality, its limited diffusion in these areas tends to limit its impact.

A plausible interpretation is that the benefits of digital infrastructure are unevenly distributed, with individuals and firms already occupying more favorable socioeconomic positions better equipped to leverage digital opportunities (Wang and Shen 2024).

regions. This pattern suggests a saturation effect, whereby economically and digitally advanced territories experience diminishing marginal impacts from broadband expansion. In contrast, southern provinces exhibit stronger effects, reflecting greater sensitivity to digital infrastructure development. Similar, albeit somewhat more heterogeneous, dynamics are observed when confining the analysis to the post-treatment period (2019–2022). This evidence is perfectly consistent with the patterns that also emerged at the macro-territorial level (see Table 8), strengthening the interpretation of a nonlinear relationship between digitalization and inequality. Additional provincial-level estimates of direct and indirect effects, controlling for socioeconomic covariates, are reported in the Supplementary Materials.

Overall, the spatial analysis complements the benchmark findings by demonstrating that the inequality-enhancing effects of broadband are not confined to individual provinces but disseminate across space. This highlights the importance of coordinated regional policy responses: in the absence of spatially integrated strategies, fragmented broadband interventions risk creating clusters of advantage and disadvantage, with distributive effects extending beyond local jurisdictions. In this sense, the spatial evidence reinforces the logic of cumulative causation logic discussed in the theoretical framework, whereby digitalization generates path-dependent divergence across territories once initial disparities take root. As inequality rises in one province, neighboring areas are drawn into similar trajectories, implying that localized policy measures may be insufficient to counteract the distributive risks of broadband expansion. Instead, effective mitigation requires supra-local coordination-aligning broadband investment with complementary policies in education, inclusion, and governance-to prevent spatial spillovers from consolidating persistent inequality traps.

7 Conclusion

This study examined the impact of the introduction and expansion of ultra-broadband infrastructure in rural areas on income inequality in Italy. These areas, classified by national and EU regulations as “market failure zones,” are therefore eligible for public intervention through dedicated investment programs, such as Law No. 55/2019 and the National Ultra-Broadband Plan.

Using a Difference-in-Differences econometric framework, the analysis assessed whether the deployment of digital infrastructure in peripheral areas after 2019 has contributed to widening income disparities.

The empirical results provide robust evidence that the introduction of ultra-broadband, particularly through FTTH and Target 100 Mbps coverage in rural areas—is associated with a statistically significant increase in income inequality across provinces. This finding holds across multiple inequality indicators, including the Gini, Theil, and Atkinson indices, underscoring the robustness of the result. Taken together, the evidence indicates that digital expansion alone is insufficient to generate equitable outcomes. The average effect of broadband deployment appears to disproportionately benefit those already in more favorable socioeconomic positions, amplifying pre-existing disparities.

When territorial heterogeneity is taken into account, however, a more nuanced picture emerges. In Southern Italy, where structural weaknesses and lower baseline development prevail, broadband expansion tends to exacerbate inequality. By contrast, in northern regions with stronger economic and digital foundations, broadband infrastructure can contribute to narrowing disparities once a critical coverage threshold is reached. In remote and inner provinces, the results are heterogeneous; inequality effects are more pronounced in areas characterized by higher marginalization. This pattern reflects the dual nature of the digital divide, whereby technological and socioeconomic barriers can either reinforce or mitigate inequality, depending on the local context.

The spatial analysis confirms the existence of significant geographical interdependencies. Inequality in a given province is influenced not only by its own level of broadband coverage but also by the coverage in neighboring areas. Such spatial spillovers suggest that the effects of digital infrastructure transcend administrative boundaries, emphasizing the need for coordinated regional planning. Disparities in broadband coverage across adjacent territories can undermine the effectiveness of localized broadband investments, while uncoordinated network externalities may deepen regional divides.

These findings are consistent with the theoretical framework proposed in this study. Broadband expansion influences inequality through three principal channels: asymmetric productivity gains, uneven structural transformation, and unequal redistribution of digital benefits. In contexts where adequate institutional and human capital capacity are absent, these channels interact cumulatively, producing path-dependent trajectories of divergence. Hence, digitalization alone cannot guarantee convergence without strong complementary policies that strengthen local absorptive capacity and address territorial polarization.

From a policy perspective, the results convey important implications for the design and governance of digital infrastructure strategies. They indicate that broadband expansion must be complemented by integrated interventions addressing the structural, institutional, and spatial mechanisms underlying inequality.

First, policies should explicitly target asymmetric productivity effects. Investments in rural and peripheral areas must be accompanied by programs that strengthen local digital capabilities—particularly among small firms, self-employed workers, and low-income households.

Second, to counter structural polarization, broadband policy should be integrated with regional industrial and innovation strategies. Investments should prioritize sectors capable of locally absorbing digital technologies. Alignment with Smart Specialization Strategies and regional innovation ecosystems can enhance the transformative potential of broadband while preventing the concentration of value-added activities in already advanced urban areas.

Third, redistributive mechanisms should be incorporated into digital policy to mitigate exclusion among vulnerable groups, especially in disadvantaged and inner areas. Such financial and social instruments are essential to transform digital connectivity into inclusive economic opportunities.

Fourth, given the strong spatial spillovers documented in the analysis, digital governance should operate at a supra-local scale. Coordinated planning across

BRIEF COMMUNICATION

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Socioeconomic factors influencing rural-urban ambulance response time disparities in Connecticut

Eashwar Krishna^{1*}

Abstract

Across the U.S, it is a documented fact that rural areas have longer ambulance response times and tend to have lower median income. The objective of this study was to test if the rural-urban emergency medical service (EMS) response time disparity was related to wealth disparity in the state of Connecticut. All mean EMS response times were sourced from the 2016 Office of Emergency Medical Services Data Report. Rural definitions were sourced from the Connecticut Office of Rural Health. Median income data was drawn from the Connecticut Office of Policy and Management. A Mann-Whitney U test determined if the average rural EMS response time was greater than the non-rural EMS response time. Pearson coefficients quantified the relationship between median income and EMS response time. A t-test ascertained if the average median income differed between the two datasets. The mean EMS response time was 12.98 min (SD = 3.36) rural and 8.26 min (SD = 2.12) non-rural. Rural mean response time and median income were not significantly correlated ($r = -.148$, $p = .247$); non-rural mean response time and median income were also not significantly related. No significant disparity was detected ($t = 0.478$, $p = .633$) between the mean rural household income (\$98,258) and mean non-rural household income (\$95,706). Significant disparities in EMS response times can exist between rural and non-rural towns separate from median income trends, as is the case in Connecticut. These findings may have limited generalizability because of Connecticut's relatively high median income as compared to other states yet may be relevant to states with similar economic metrics.

Keywords Rural health, Emergency medical services, Ambulances, Connecticut, Public Health

Introduction

It has been demonstrated that shorter ambulance response times improve immediate and long-term outcomes during cardiac arrests [1, 2]; for example, O'Keefe et al. [3] found for every one minute reduced in response time, the chance of survival increased by 24%. While patient survival rates of an adverse cardiac event can be greatly improved through bystander education and

intervention, the same cannot be said for other out-of-hospital medical emergencies. One such case is foreign body airway obstruction events: in only about 5–13% of cases are bystanders able to make a correct diagnosis and perform the appropriate maneuver; moreover, approximately 30–40% of all such events occur when the victim is alone [4]. On average, the victim has six minutes before they begin to suffer from hypoxia-related brain damage.

As such, it is a public health imperative to provide timely professional care to the acutely ill and injured. Mell et al. [5] report that across the USA, rural response times are almost double suburban and urban responses. A good amount of heterogeneity among the results of

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rural-urban response time disparity studies has also been documented [6], creating the need to study various regions of the United States in isolation from each other.

While the relationship between median income and response times [7], as well as rurality and response times [5], has been studied, the interrelation of these two factors has not yet been the subject of sufficient research. Connecticut, a northeastern state of the USA, is an excellent case study to understand the unique challenges that face rural EMS agencies and the role, if any, of median income. On a national level, urban counties tend to have higher median income than rural counties [8]; in contrast, Connecticut has the highest median income and lowest poverty rates for rural areas in the country [9], and possibly no significant rural-urban wealth disparity.

I hypothesized that a significant disparity between rural and non-rural response times will persist in Connecticut; however, I expected that this disparity would not be correlated with median income.

Methods

All mean EMS response times were sourced from the 2016 Office of Emergency Medical Services Data Report, published by the Connecticut Department of Public Health. Towns, neighborhoods, or villages with fewer than thirty recorded EMS encounters, incomplete ZIP (Zone Improvement) code data, or incomplete time point data were excluded from the report [10]. It should be noted that many of these excluded areas were in fact neighborhoods under the jurisdiction of a recorded town. The only independent rural town without a mean response time reported due to insufficient reports was Lyme; Union, North Canaan, Sharon, and Warren were excluded from the report with no explicit explanation yet it may be assumed that they lacked complete time points or ZIP code data. Stafford, Griswold, and Killingly – classified for the purposes of this study as non-rural – were also excluded from the report due to incomplete data.

The Connecticut Office of Rural Health has published a list of towns considered rural according to the state definition, which was adopted for use in this study; other rural definitions such as RUCC and RUCA were not applicable to this study as national standards do not use towns as their fundamental units of study. Towns were chosen as the focus of this study because they are the smallest units of governance in Connecticut, and as such EMS policy and median income can vary widely between even neighboring towns in the same county.

All towns with a population of fewer than ten thousand residents as well as a population density of five hundred people or fewer per square mile were classified as rural [11], yielding 68 rural towns in total; 63 towns were used for data analysis due to the remaining five lacking data. Urban and suburban towns were grouped together since

the nationwide mean times for both groups were highly similar; in total, 95 towns and cities were used for the non-rural comparison group. The dataset used in the analysis can be found in Appendix A and B.

Average household median income data was drawn from the Municipal Fiscal Indicators dataset released by the Connecticut Office of Policy and Management [12]. Median income and mean response time data were paired for each town, and the overall dataset was split into rural and non-rural sets. Only towns being used for the response time comparison were included in these datasets; the eight towns excluded for the aforementioned reasons were also excluded from the median income calculations. The median income sets can be found in Appendix C and D.

Statistical analysis

The null hypothesis was that the distribution of mean response times (MRT) for rural and urban towns would be equal, whereas the alternative hypothesis was that the distributions would not be equal. A two-sample Wilcoxon-Mann-Whitney U test with $(\alpha)=0.05$ was performed to test if the rural distribution of response times was stochastically different than the urban distribution. The two datasets had different distribution shapes, and thus a non-parametric test was necessary. Measures of central tendency were calculated and the Tukey fence method was used to better describe the data and discover outliers, respectively.

Pearson coefficients of correlation were calculated for the relationship between average household median income and mean response time for both sets. Furthermore, a two-tailed t-test with $(\alpha)=0.05$ was performed to determine if median household income varied between the rural and non-rural datasets. The data were analyzed using IBM SPSS statistics (Version 29).

Ethics approval

This study only analyzed publicly-available, de-identified data and as such did not meet the criteria for IRB review according to 45 CFR 46.102.

Results

A total of 158 towns were included in the analysis. The mean response time for the rural dataset was 12.98 min ($SD=3.6$), while the same for the non-rural data was 8.26 min ($SD=2.12$); the distributions in the two groups differed significantly ($z=8.3464$, $p<.001$). Median response times were 12.8 min and 7.7 min for rural and non-rural respectively. The observed standardized effect size was medium at 0.66, meaning that the magnitude of the difference between the two datasets was sizable. No outliers were observed in the rural data; Simsbury and Weston were non-rural dataset outliers with 15.8- and

14.6-minute mean response times, respectively, yet were still included in the analysis.

Neither rural mean response time and median income ($r = -.148$, $p = .247$) nor non-rural mean response time and median income ($r = .137$, $p = .185$) were found to be significantly correlated. There was no significant difference in average household median incomes between the two datasets ($t = 0.478$, $p = .633$).

Discussion

While rural response times statewide are lower than the national rural average, a significant disparity still exists between non-rural and rural towns within Connecticut; this inequity has serious implications for patient outcomes. Though only a few urban/suburban towns have a mean response time below five minutes, generally considered the gold standard, the average non-rural MRT is not significantly far from it. In contrast, the average rural response time in Connecticut is almost thirteen minutes, well over double the time necessary to consistently save victims of traumatic injuries, choking, or other severe events.

In addition, the data makes it evident that median income has no statistically significant impact on response time within either dataset. Moreover, there is no significant wealth disparity as measured by household median income between rural and non-rural areas. This suggests that more focus should be directed towards other aspects of rural EMS, such as staffing shortages and town policies, rather than solely focusing on socioeconomic factors. It should be noted that this conclusion is specific to Connecticut; it may or may not be the case elsewhere that the most pertinent issue to response time disparity is a rural-urban wealth disparity. While lower-income towns may struggle with EMS procurement and operational costs, which can be unpredictable [13], more than higher-income towns, multiple papers have demonstrated that the impact of non-financial factors may be more influential in resolving the response time disparity. For example, Wei Lam et al. found that changes to dispatch policy and more effective allocation of existing ambulances would be more effective than merely adding more ambulances in reducing response times overall [14]. As for rural response times specifically, poor access to primary care for non-acute health management, provider burnout, and lack of opportunities for provider-initiated workplace improvements have been cited by Newton et al. as issues contributing to lengthy rural EMS response times [15]. The same study indicated that directing federal and state funding to community paramedicine initiatives may be effective at managing the underlying health disparities present in rural and remote communities; failure to provide consistent primary care to these populations tends

to result in increased EMS utilization [16], which can put already-strained units out-of-service.

Other communities may find it beneficial to conduct small-scale, local investigations regarding EMS response in their jurisdiction. Much of the existing work on response time and rurality focuses on county [17] or ZIP codes [5] as geographic units of analysis; however, this is frequently not the level at which policy and quality improvement interventions are implemented. For example, in the New England region of the United States, which includes Connecticut, municipalities such as villages, towns, and cities regulate emergency services independently, with limited oversight from the state. The implication of this fact is that county-focused analysis may be erroneously grouping together a vastly heterogeneous number of local EMS services, with varying degrees of funding, privatization, and quality improvement initiatives. Conversely, ZIP code-focused analysis may become too granular, as a single municipality frequently covers multiple ZIP codes. This study attempts to illustrate municipality-level differences, which emphasize the necessity of local policy innovations to combat long response times.

Such policy changes are already underway in Connecticut, and this study underscores their necessity and potential impact; since rural-urban response time disparities in the state do not appear to be correlated with median income, such disparities are more remediable by tailoring solutions to the unique challenges faced by rural communities. Rather than utilize limited funds to solely subsidize equipment purchases, it may be far more impactful to pursue EMS innovation and fundamental alterations to the existing pre-hospital care infrastructure. A prime example of such a project is the Mobile Integrated Health workgroup, chaired by the Connecticut Department of Public Health, which aims to create a community paramedicine initiative that will integrate with existing EMS systems. Notably, the group's most recent report indicates a goal to "triage and connect non-urgent 9-1-1 callers with relevant caregivers and assistance instead of dispatching an ambulance crew" [18]. The results of this study in combination with existing research indicate that it will be efficacious for additional attention to be given to community-level interventions in rural areas such as health literacy programs, increased incentives for primary care providers, and increased expansion of mobile integrated health solutions in Connecticut.

Limitations

In order to accurately capture the effect of median income and town policy, analysis was conducted by town lines; however, the source data for mean response time did not classify towns by strict jurisdictional borders, and as such included neighborhoods or villages of towns

as separate entries. In these situations, the town center data point was used for mean response time; as such, the fidelity of the data is a limitation of this research. This study also limited its analysis of socioeconomic factors to median household income, and other such factors may affect response times.

The data utilized in this study, collected in 2016, may present certain implications for the results. The age of the data is a critical factor to consider, as it may not fully capture recent changes in EMS infrastructure, technological advancements, or policy shifts that have occurred since then. For instance, while rural towns have predominantly relied on volunteer services for coverage, a declining number of volunteers has led some of these towns to contract private agencies. Subsequently, these municipalities reported decreased response times [19]. Such trends, if significant across most towns included in the rural dataset, may be hypothesized to decrease the disparity in response time observed in this study.

Despite these shifts, a more recent report from the Department of Public Health (DPH) indicates that the overall environment of EMS as it relates to rural-urban response time disparities has remained roughly the same. At the time of the analysis, the non-rural mean response time (MRT) was 8.26 min; this report was published five years later and reports a suburban MRT of 8.12 min and an urban MRT of 7.39 min [20]. Rural towns were noted to have a mean response time of 10.07 min in the more recent report, compared to 12.8 min in this study. This disparity can be explained by data reporting standards changing; namely, at the time of the data analysis in this study, median response times were not stratified by rural status by the DPH. In the more recent report, several towns that were previously under the non-rural dataset have been included in the rural MRT average, which understandably lowers this number. While this phenomenon is partially explained by population trends and possibly differences in methodology, a repetition of the analysis in this study with newer data would be warranted if town-specific conclusions need to be drawn. However, the general trend illustrated by this study and the interpretation thereof remains relevant to both the state of Connecticut as well as other communities who might benefit from investigating novel solutions to the indisputable disparity in rural pre-hospital care.

Conclusion

There exists a significant rural-urban disparity in ambulance response times in Connecticut which is decoupled from median income, a concerning trend that demands amelioration in the near future to ensure equitable access to timely healthcare. The most effective strategy to do so will be one that focused on policy and innovation in both

emergency medical services as well as care focusing on chronic and preventable conditions.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1007/s43999-024-00055-9>.

Supplementary Material 1

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Data availability

Data was publicly available through the Connecticut Department of Public Health.

Declarations

Ethical approval and consent to participate

This study did not qualify to be reviewed by an IRB.

Conflict of interest

The author reports no conflict of interest for this study.

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ivcrc: An Instrumental Variables Estimator for the Correlated Random Coefficients Model

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ivcrc: An Instrumental Variables Estimator for the Correlated Random Coefficients Model

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March 3 2022

Abstract

We discuss the **ivcrc** module, which implements an instrumental variables (IV) estimator for the linear correlated random coefficients (CRC) model. The CRC model is a natural generalization of the standard linear IV model that allows for endogenous, multivalued treatments and unobserved heterogeneity in treatment effects. The estimator implemented by **ivcrc** uses recent semiparametric identification results that allow for flexible functional forms and permit instruments that may be binary, discrete, or continuous. The **ivcrc** module also allows for the estimation of varying coefficients regressions, which are closely related in structure to the proposed IV estimator. We illustrate use of **ivcrc** by estimating the returns to education in the National Longitudinal Survey of Young Men.

Keywords: **ivregress**, Instrumental Variables, Correlated Random Coefficients, Heterogeneous Treatment Effects, Varying Coefficient Models, Returns to Schooling

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1 Introduction

In this paper we describe the `ivcrc` module for Stata, which implements a linear instrumental variables (IV) estimator for the correlated random coefficients (CRC) model. The CRC model relaxes the constant effects assumption of the standard linear IV model by allowing for random coefficients, which capture unobserved heterogeneity in the causal effect of the endogenous variables, X , on the outcome variable, Y . Heckman and Vytlacil (1998) and Wooldridge (1997, 2003, 2008) showed that if there is no unobserved heterogeneity in the way the instrument, Z , affects the endogenous variables, X , then the usual linear IV estimator (e.g., `ivregress`) will estimate the mean of the random coefficients. However, this assumption is uncomfortably asymmetric: treatment effects can be unobservably heterogeneous, but instrument effects cannot.

Masten and Torgovitsky (2016) address this drawback by showing how to allow for heterogeneity in the relationship between Z and X . The identification arguments developed there allow for the instrument to be binary, discrete, or continuous, and depend only on familiar, low-level assumptions about exclusion, exogeneity, and instrument relevance. The arguments also lead to a semiparametric estimator that does not suffer from the curse of dimensionality. In Section 2, we briefly review the identification results and estimation approach developed in Masten and Torgovitsky (2014, 2016). The `ivcrc` module implements their proposed estimator.

The structure of the estimator turns out to be quite similar to a common estimator for the varying coefficient models (e.g., Fan and Zhang 2008; Park, Mammen, Lee, and Lee 2015). We have written `ivcrc` to include a standard estimator for these models as a special case. We briefly describe varying coefficient models in Section 3.

In Section 4 we discuss syntax and options for the `ivcrc` module. In Section 5 we illustrate the module by estimating the return to schooling with a widely-used extract from the National Longitudinal Survey of Young Men. For further examples, see Gollin and Udry (2020), who used the `ivcrc` module to estimate agricultural production functions, Morales-Mosquera (2019), who used it to estimate the effects of police infrastructure in Colombia, and Masten and Torgovitsky (2014), who used the estimator to revisit Chay and Greenstone’s (2005) analysis of the effect of air pollution on housing prices.

2 The Correlated Random Coefficients Model

2.1 Model and Motivation

The simplest form of the model estimated by `ivcrc` has the outcome equation

$$Y = B_0 + B_1X, \tag{1}$$

where Y is an observed outcome, X is an observed explanatory variable, and both B_0 and B_1 are unobserved random variables. The model is described as a random coefficients model due to the treatment of B_1 as an unobserved random variable. Economists have long been interested in such models (Wald 1947; Hurwicz 1950; Rubin 1950; Becker and Chiswick 1966). To allow for endogeneity, X is permitted to be arbitrarily dependent with both B_0 and B_1 . This feature makes the model one of *correlated* random coefficients.¹

It is helpful to compare (1) with the outcome equation for the textbook linear model:

$$Y = \alpha + \beta X + U, \quad (2)$$

where α and β are fixed (deterministic) parameters, and U is an unobservable random variable with mean zero. This model also allows for endogeneity by permitting X to be dependent with U . The distinction between U in (2) and B_0 in (1) is not important, since one can view B_0 as being equal to $\alpha + U$. Rather, the important difference between (2) and (1) is that the coefficient on X in (2), i.e. β , is deterministic, whereas the coefficient on X in (1), i.e. B_1 , is a random variable. The interpretation is that in (2) the causal effect of X on Y is the same for all agents, whereas in (1) it is a random variable that can be dependent with X . This important difference allows for heterogeneous treatment effects and selection on the gain of the sort described in Heckman, Urzua, and Vytlacil (2006b). One can view (2) as a special case of (1) with a degenerate B_1 .

Textbook discussions of (2) show that β is identified if there exists an instrument Z such that $\text{Cov}(U, Z) = 0$ and $\text{Cov}(X, Z) \neq 0$. The corresponding IV estimator can be implemented in Stata with the `ivregress` command. However, if the data is in fact generated by (1), then this estimator converges to

$$\frac{\text{Cov}(Y, Z)}{\text{Cov}(X, Z)} = \mathbb{E} \left[B_1 \times \frac{X(Z - \mathbb{E}(Z))}{\mathbb{E}[X(Z - \mathbb{E}(Z))]} \right]. \quad (3)$$

This quantity is difficult to interpret in general (Garen 1984; Wooldridge 1997; Heckman and Vytlacil 1998). It is a weighted average of the causal effect of X on Y ; that is, a weighted average of B_1 . The weights, however, can be both positive and negative. It generally does not equal the unweighted average of B_1 unless B_1 is independent of (X, Z) , which would rule out the type of selection on the gain scenario described in Heckman et al. (2006b).

A natural question is whether there are additional assumptions under which the IV estimator provided by `ivregress` would consistently estimate a parameter that is easier to interpret. For example, are there additional conditions under which this estimator converges to the average partial effect, $\mathbb{E}[B_1]$? Heckman and Vytlacil (1998) and Wooldridge (1997, 2003, 2008) show that there are

¹ This terminology seems to have been first used by Heckman and Vytlacil (1998). In earlier work, some authors, for example, Conway and Kniesner (1991), had used the adjective “correlated” to describe an unrestricted correlation structure between the random coefficients on different explanatory variables. Our model also allows for this.

indeed such conditions, namely, the assumption that the causal effect of Z on X is homogenous. While convenient, this type of homogeneity assumption is uncomfortably asymmetric. It enables the additional heterogeneity in equation (1) relative to equation (2) only by assuming away the same type of heterogeneity in the analogous relationship between Z and X .²

2.2 Identification and Estimation by Conditional Linear Regression

Given these negative results, it is worthwhile to consider estimators other than `ivregress`. The `ivcrc` module provides such an estimator. This estimator is based on the following control function argument, which is developed more formally in Masten and Torgovitsky (2016).³

Suppose that there exists an observable variable R such that $X \perp\!\!\!\perp (B_0, B_1) \mid R$. The variable R is a “control function” (or sometimes, and more loosely, a “control variable”) because it controls for the endogeneity in X . That is, while X is endogenous in the sense of being unconditionally dependent with (B_0, B_1) , it is exogenous after conditioning on the control function R . In practice, R is constructed from the instrument; we explain the derivation and construction of R in more detail in Section 2.3.

Given the availability of a variable R with this property, it is straightforward to see that one could consistently estimate the vector $\beta(r) \equiv \mathbb{E}[B \mid R = r]$ where $B \equiv [B_0, B_1]'$ by a linear regression of Y on X *conditional* on $R = r$. Letting $W \equiv [1, X]'$ so that $Y = W'B$, one has

$$\mathbb{E}[WW' \mid R = r]^{-1} \mathbb{E}[WY \mid R = r] = \mathbb{E}[WW' \mid R = r]^{-1} \mathbb{E}[WW'B \mid R = r] = \beta(r), \quad (4)$$

where the second equality uses the assumption that B is independent of X (and hence W), conditional on R . In order for this argument to work, it must be the case that $\mathbb{E}[WW' \mid R = r]$ is invertible, which is the usual condition of no perfect multicollinearity, but now conditional on $R = r$. Intuitively, there must still be some variation left in X after conditioning on $R = r$. Assuming that this is the case for all r in the support of R , one can average up the linear regression estimands on the right-hand side of (4) to obtain $\mathbb{E}[B] \equiv \mathbb{E}[\beta(R)]$, and hence the average partial effect of X on Y , i.e. $\mathbb{E}[B_1]$.

This identification argument suggests an estimator given by an average of conditional ordinary least squares (OLS) estimators. The conditioning is incorporated by applying kernel weights to each observation, where the weights reflect the distance of R from r . More concretely, given a

² An influential literature started by Imbens and Angrist (1994), Angrist and Imbens (1995), Angrist, Imbens, and Rubin (1996), and Angrist, Graddy, and Imbens (2000) has provided conditions under which the IV estimator provided by `ivregress` can be interpreted as a local average treatment effect (LATE) or a weighted average of various LATEs. While related, these arguments are nonparametric, and in particular do not use the linearity in X of the CRC model.

³ This paper builds on a large literature on control functions, including Heckman (1979), Heckman and Robb (1985), Smith and Blundell (1986), Blundell and Powell (2004), Florens, Heckman, Meghir, and Vytlačil (2008), and Imbens and Newey (2009), among many others.

sample $\{Y_i, X_i, R_i\}_{i=1}^n$, a conditional regression estimator of Y on W near $R = r$ is given by

$$\hat{\beta}(r) \equiv \left(\sum_{i=1}^n k_i^h(r) W_i W_i' \right)^{-1} \left(\sum_{i=1}^n k_i^h(r) W_i Y_i \right), \quad (5)$$

where $k_i^h(r) \equiv h^{-1} K((R_i - r)/h)$ with K a second-order kernel function and $h > 0$ a bandwidth parameter.

The conditional OLS estimator (5) displays the same type of bias-variance tradeoff that is familiar from nonparametric kernel regression. As $h \rightarrow \infty$, $k_i^h(r) \rightarrow K(0)$ for all i , so that $\hat{\beta}(r)$ is just the estimator from a usual linear regression of Y on W . We expect this estimator to be biased for $\mathbb{E}[B]$ if X is endogenous. Given the control function assumption, this bias disappears as $h \rightarrow 0$, but at the cost of higher variance in using fewer effective observations in computing $\hat{\beta}(r)$. Balancing these two concerns entails using fewer than n effective observations, and as a consequence $\hat{\beta}(r)$ will have a slower-than-parametric rate of convergence for $\beta(r)$.

As a parameter of interest, $\beta(r)$ has a clear interpretation as the average partial effect of X on Y , conditional on $R = r$. Variation in this parameter as a function of r indicates treatment effect heterogeneity. We can average $\beta(R)$ for R in some known set $\mathcal{R}$ to obtain the average partial effect for the subpopulation with $R \in \mathcal{R}$. A natural estimator of this average is given by

$$\hat{\beta}_{\mathcal{R}} = \frac{\sum_{i=1}^n \hat{\beta}(R_i) \mathbb{1}[R_i \in \mathcal{R}]}{\sum_{i=1}^n \mathbb{1}[R_i \in \mathcal{R}]}, \quad (6)$$

where $\mathbb{1}[\cdot]$ is the indicator function that is 1 if $\cdot$ is true and 0 otherwise. At least in principle, $\hat{\beta}_{\mathcal{R}}$ can be estimated at the parametric $\sqrt{n}$ rate (see Masten and Torgovitsky 2014, or, for a more general discussion, Newey 1994). If the local Gram matrix is invertible for (almost) every r in the support of R , then $\mathcal{R}$ can be taken to be the entire support of R , so that (6) becomes an estimator of the unconditional average of B .

A more general version of (1) is

$$Y = B_0 + \sum_{j=1}^{d_x} B_j X_j + \sum_{j=1}^{d_1} B_{d_x+j} Z_{1j} \equiv W' B, \quad (7)$$

where X is now a d_x -dimensional vector of potentially endogenous explanatory variables and $Z_1 \in \mathbb{R}^{d_1}$ is a vector of exogenous explanatory variables. For notation, we combine these variables and their coefficients together with the constant term as $W \equiv [1, X', Z_1']'$ and B . We rename the excluded exogenous variable as Z_2 , and combine the exogenous variables (included and excluded instruments) together into a vector $Z = [Z_1', Z_2']'$. The required condition on the control function is now that $W \perp\!\!\!\perp B \mid R$, so that both X and Z_1 are exogenous after conditioning on R . Given this condition, the identification argument (4) and the estimators (5) and (6) follow exactly as before.

2.3 Estimation of the Control Function

We have shown how a control function, R , can be used to estimate interesting parameters in a CRC model, but we have not yet explained how one can find or construct such a control function. The most common approach is to assume that for each $j = 1, \dots, d_x$, there exists a function h_j and unobservables $V \equiv [V_1, \dots, V_{d_x}]' \in \mathbb{R}^{d_x}$ such that

$$X_j = h_j(Z, V_j) \quad \text{for each } j, \quad (8)$$

where $h_j(z, \cdot)$ is strictly increasing for each z . As shown by Imbens and Newey (2009) and Masten and Torgovitsky (2016), if $(B, V) \perp\!\!\!\perp Z$, then $R \equiv [R_1, \dots, R_{d_x}]'$ is a valid control function, where $R_j \equiv F_{X_j|Z}(X_j | Z)$ and $F_{X_j|Z}(x_j | z) \equiv \mathbb{P}[X_j \leq x_j | Z = z]$ is the population conditional distribution function of X_j , given Z . The components R_j of this control function can be interpreted as providing the conditional rank (relative position) of X_j given Z . The `ivcrc` module is written primarily with this choice of control function in mind, although the user can provide a different choice if desired. In that case, the estimator can be viewed as estimating the varying coefficient model discussed in the next section.

We refer to Masten and Torgovitsky (2016) for more theoretical details on the interpretation and restrictiveness of maintaining (8); see also Chernozhukov and Hansen (2005) and Torgovitsky (2015). Here we focus on the implications for implementing (4) and (5) with R as the resulting conditional ranks. The first implication is that it may be useful to make a distinction between different components of the endogenous variables, X . For example, if X_2 is a deterministic transformation of X_1 , say $X_2 = X_1^2$, then X_2 is also fully determined by R_1 . As a result, there is no need to separately estimate and condition on R_2 . In the terminology of Masten and Torgovitsky (2016), X_1 is a basic endogenous variable, and $X_2 = X_1^2$ is a derived endogenous variable.

Derived endogenous variables require special treatment, since they appear as part of the vector of explanatory variables W , but are not included as part of the conditioning variables Z in the definition of $R_j \equiv F_{X_j|Z}(X_j | Z)$. More formally, a component X_j of X is a derived endogenous variable if it can be written as $X_j = g_j(X_{-j}, Z)$ for some known function g_j . Interaction terms and other nonlinear functions form the primary examples of derived endogenous variables. The `ivcrc` module handles derived endogenous variables using the `dendog` option discussed in Section 4. The empirical illustration in Section 5 provides an example of its use.

A second issue raised by this choice of R is that it is not directly observed in the data. Instead, we need to estimate $R_{ji} = F_{X_j|Z}(X_{ji} | Z_i)$ in a first step for each basic endogenous variable X_j and each observation i . The `ivcrc` module approaches this problem by estimating conditional quantile functions and then inverting them using the pre-rearrangement operator studied by Chernozhukov, Fernandez-Val, and Galichon (2010). This operator translates an estimator of a conditional quantile function, say $\hat{Q}_{X_j|Z}(\cdot | z)$, into an estimator of a conditional distribution function through the

relationship

$$\widehat{F}_{X_j|Z}(x_j | z) = \int_0^1 \mathbb{1} \left[\widehat{Q}_{X_j|Z}(s | z) \leq x_j \right] ds. \quad (9)$$

For estimating $\widehat{Q}_{X_j|Z}(s | z)$, the `ivcrc` module uses linear quantile regression (see e.g. Koenker, 2005) as implemented by Stata’s built-in `qreg` command. The generated regressors $\{\widehat{R}_{ji}\}_{i=1}^n$ are then constructed by substituting (X_{ji}, Z_i) into (9) for every i .

A third point that arises when using this choice of R is that (6) can be simplified when there is only one basic endogenous variable. This is because $R \equiv F_{X|Z}(X | Z)$ is uniformly distributed when X is continuous. As a result, the probability that R lands in any region $\mathcal{R}$ is known a priori and does not need to be estimated. The population average of $\beta(R)$, conditional on $R \in \mathcal{R}$ in this case reduces to

$$\beta_{\mathcal{R}} = \lambda(\mathcal{R})^{-1} \int_{\mathcal{R}} \beta(r) dr, \quad (10)$$

where $\lambda(\mathcal{R})$ is the Lebesgue measure of the set $\mathcal{R}$. When equation (10) holds, `ivcrc` estimates it by substituting the (known) value of $\lambda(\mathcal{R})$ and numerically approximating the integral $\int_{\mathcal{R}} \widehat{\beta}(r) dr$ that replaces $\beta(r)$ with $\widehat{\beta}(r)$.

A fourth point that is worth reemphasizing is that in order for (4) to exist, the Gram matrix $\mathbb{E}[WW' | R = r]$ must be invertible. That is, there must not be perfect multicollinearity among the regressors after conditioning on $R = r$. When using the conditional rank for R , conditioning on $R = r$ still leaves variation in the basic endogenous variables as long as the excluded instrument, Z_2 , is appropriately dependent with X near its r th quantile. See Masten and Torgovitsky (2014, 2016) for a more detailed discussion of this point. A consequence for implementation is that it is necessary to exclude from $\mathcal{R}$ regions over which this instrument relevance condition fails.

2.4 Bandwidth Selection

The `ivcrc` module implements automated bandwidth selection for h based on a rule-of-thumb method proposed by Fan and Gijbels (1996). We first estimate $\mathbb{E}[Y | R = r, X = x]$ as a fourth order polynomial in r , linearly interacted with x . We use this regression to produce estimates of the second derivative of $\mathbb{E}[Y | R = r, X = x]$ with respect to r , which we denote by $\widehat{\delta}(r)x$. The rule-of-thumb bandwidth is then defined as

$$h_{\text{ROT}} \equiv 0.58 \left(\frac{\widehat{\sigma}^2}{\sum_{i=1}^n \frac{1}{2} \widehat{\delta}(\widehat{R}_i) X_i} \right)^{1/5},$$

where $\widehat{\sigma}^2$ is the homoskedastic error variance from the polynomial regression and the constant 0.58 is derived from the roughness for single-peaked kernels. See the discussion in Hansen (2021, Section

The syntax for the `ivcrc` module is

```
ivcrc depvar [varlist1] (varlistedg = varlist2) [if] [in] [, options]
```

In terms of the IV model discussed in Section 2, `depvar` is Y , `varlist1` consists of the components in Z_1 , `varlistedg` are the basic endogenous variable components of X , and `varlist2` are the components in Z_2 . The required components of the syntax are `depvar`, `varlist2`, and `varlistedg`, while the remaining terms in brackets are optional.

The module allows for the options shown in Table 1. The `dendog` option allows the user to specify a list of endogenous variables that should be treated as derived (rather than basic), with the implications for implementation discussed in Section 2. The `bootstrap` option controls the calculation of standard errors and confidence intervals. Note that `ivcrc` does not compute these by default, because the bootstrap procedure can be computationally intensive. The `kernel` and `bandwidth` options allow the user to change the kernel function K and bandwidth h used to compute the weights in (5). If the input for `bandwidth` is a list of numbers (separated by commas), then `ivcrc` will compute different estimates for each bandwidth. The computational efficiency of specifying several bandwidths at once is especially useful when calling `bootstrap` for standard errors and confidence intervals. The `ranks` option controls the degree of accuracy for approximating the integral in (9).

The `average` option determines the set $\mathcal{R}$ over which the local estimates $\hat{\beta}(r)$ are averaged and controls how this averaging is implemented. For example, `average(.1(0).3)` sets $\mathcal{R} = [.1, .3]$ and uses the empirical mean to evaluate the integral in (10). The module interprets a grid step of 0 as a request for computing $\hat{\beta}_{\mathcal{R}}$ using the sample averaging formula (6) that does not use knowledge of the distribution of R . Alternatively, specifying `average(.1(.01).3)` sets $\mathcal{R} = [.1, .3]$ and uses grid steps of .01 to numerically evaluate the integral. Multiple non-overlapping sets can be specified by adding commas. If the `report` suboption is given, then estimates on each set will be reported separately together with the overall estimate. For example, `average(.1(0).3, .5(0).8, report)` would report the estimate of $\beta_{\mathcal{R}}$ just discussed, along with another empirical average estimate for $\mathcal{R} = [.5, .8]$. The grid method supports the `report` suboption as well.

There are two situations in which the module will always use (6) instead of attempting to numerically integrate (10). The first is when there is more than one basic endogenous variable, in which case R is a vector with a joint distribution that is not known a priori and (10) is not valid. If a user specifies a list of subsets `average(lb1(g1)ub1, ..., lbN(gN)ubN)` when there are multiple basic endogenous variables, the module interprets each subset `lbn(gn)ubn` as belonging to the n th endogenous variable in order of appearance in `varlistedg`. Due to the difficulty of specifying sets in higher dimensions, more general multidimensional subset estimates may be obtained either by permuting this syntax, or by storing the local estimates $\hat{\beta}(r)$ using the `savecoef` option and subsequently computing any desired subset average. This is not essential to the method, but

allowing for more general specifications would complicate the syntax significantly without providing much in the way of useful flexibility.

The second case in which `ivcrc` only uses the empirical average (6) is when the `varcoef` option is called. Passing `varcoef(varlist)` skips the estimation of $\hat{R}_i$ and uses the variables in `varlist` in its place. Since the density of these variables is generally not known a priori, (10) may not be true, so (6) is used. The average in (6) can still be taken over some specified subset $\mathcal{R}$, and such a set is still specified using the `average(lb(0)ub)` syntax. Note that using both the ($varlist_{edg} = varlist_2$) syntax and passing `varcoef` as an option will generate an error.

5 Using ivcrc to Estimate the Returns to Schooling

In this section, we use the `ivcrc` module to estimate the returns to schooling. Our discussion builds off of Card (1994, 2001) and Heckman and Vytlačil (1998), who note that a simple model of optimal schooling decisions (such as Becker 1975) would generate a CRC model like (1) or (7). In the notation of (7), Y would be a labor market outcome (e.g., wages) and X would be a measure of educational attainment (e.g., years of completed schooling). There are several reasons to expect confounding factors that make the direct relationship between X and Y a poor indicator of the causal effect. Some of these determinants, like family background characteristics, can be observed in data and controlled for. However, the fact that individuals have some choice over their education also suggests that some important confounding factors are inherently unobservable.

Instrumental variable strategies have been widely used to tackle this self-selection problem. Some instruments that have been used are compulsory schooling laws (Angrist and Krueger 1991; Oreopoulos 2006), the distance a teenager lives from a college (Card 1993; Mountjoy 2019), and local labor market conditions (Cameron and Heckman 1998).⁴ The argument underlying these strategies is that the proposed instrument affects an individual’s educational attainment by shifting the costs and/or benefits involved, but does not itself directly affect labor market outcomes and is uncorrelated with any other factors that do. The CRC model discussed in Section 2 allows for the effect that an instrument has on an individual’s educational choice to be correlated with their returns to schooling. So, for example, those whose education choice is affected by distance might have systematically different returns to schooling than those whose choice is not.

Our analysis uses the same data and regression specification as Card (1993) and Kling (2001), which is available as part of Cameron and Trivedi’s (2009) textbook on Stata for Microeconometrics. The data is an extract from the National Longitudinal Survey of Young Men (NLSYM) that consists of 3,010 men who were aged 24–34 in 1966. The extract contains variables from both 1966 and a follow-up survey in 1976. The data, as well as the code for the following analysis, is available at <https://github.com/a-torgovitsky/ivcrc>.

⁴ See also Carneiro, Heckman, and Vytlačil (2011) for an IV strategy that uses multiple types of instruments.

The outcome variable is `wage76`, the individual's log hourly wage in 1976. The primary endogenous variable of interest is `grade76`, the individual's highest level of schooling measured in years. The regression includes two additional endogenous variables: years of potential work experience in 1976 `exp76`, and squared experience `expsq76`. Potential work experience is defined as `exp76 = age76 - grade76 - 6`, following the standard convention for Mincer equations (Mincer 1958, 1974). To avoid perfect collinearity, we follow the literature by including experience in the outcome equation and age in the first stage equation. Since potential experience is a derived endogenous variable, this convention also allows us to illustrate the module's `dendog` option.

We include a set of sociodemographic controls for race (`black`), parent's education (`daded`, `momed`, `famed1-8`), family structure at age 14 (`momdad14`, `sinmom14`), and geographic region (`smsa66`, `smsa76`, `reg1-reg8`).⁵ While not essential to demonstrating the usage of the `ivcrc` module, the inclusion of these controls shows that the semiparametric estimator does not suffer from the curse of dimensionality. Including these controls also replicates the regression specification in Card (1993) and Kling (2001), which allows us to compare the `ivcrc` module to popular alternative estimators like 2SLS (e.g., `ivregress`).

We begin by estimating a linear regression of log wages on schooling, potential work experience, and demographic control variables. This type of regression is often referred to as a Mincer equation; see Heckman, Lochner, and Todd (2006a) for an in-depth discussion. The estimates indicate that an additional year of schooling is associated with approximately a 7.25 percent increase in 1976 wages:

```
. reg wage76 grade76 exp76 expsq76 'ControlVars', robust
```

```
Linear regression              Number of obs   =       3,010
                              F(27, 2982)       =       52.45
                              Prob > F         =       0.0000
                              R-squared        =       0.3040
                              Root MSE     =       .37191
```

```
-----+-----
              |               Robust
      wage76 |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-----+-----
      grade76 |   .0725423   .0038685    18.75  0.000   .0649572   .0801275
      ...
```

As discussed above, education is a choice variable that is likely correlated with latent factors that affect wages, even after controlling for sociodemographic characteristics. Card (1993) used an

⁵ For readability, we collect these into a local variable `ControlVars` in the `do` file for this exercise.

indicator for living (at age 14) in a county with a four-year college as an instrument for education. Proximity to a four-year college is associated with about a third of a grade higher educational attainment:

```
. reg grade76 col4 'ControlVars', robust
```

```
Linear regression               Number of obs   =       3,010
                               F(25, 2984)       =       53.50
                               Prob > F         =       0.0000
                               R-squared         =       0.2937
                               Root MSE      =       2.2591
```

```
-----+-----
               |               Robust
           grade76 |             Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-----+-----
           col4 |    .3669905   .1023706     3.58   0.000   .1662663   .5677147
           ...
```

In order for college proximity to be a valid instrument, it should have no direct effect on wages in 1976 and also be uncorrelated with other factors that are correlated with wages or schooling decisions after conditioning on observables. There are several reasons to be suspect of this requirement; see for example Kling (2001), and see Mountjoy (2019) for a modern discussion with richer geographic data. Here we simply compare estimators and take the validity of the college proximity instrument for granted.

The 2SLS estimator suggests that an additional year of schooling causes about a 13.33 percent increase in 1976 wages:

```
. ivregress 2sls wage76 (grade76 exp76 expsq76 = col4 age76 agesq76) ///
>      'ControlVars', perfect
```

```
Instrumental variables (2SLS) regression               Number of obs   =       3,010
                                                       Wald chi2(27)    =      1007.25
                                                       Prob > chi2      =       0.0000
                                                       R-squared        =       0.2030
                                                       Root MSE        =       .39614
```

```
-----+-----
           wage76 |             Coef.   Std. Err.      z    P>|z|     [95% Conf. Interval]
```



```
-----+-----
      grade76 |   .1333034   .0493359   2.70   0.007   .0366068   .2300001
...

```

This interpretation presumes that the causal effect of schooling on wages is constant. It yields the potentially puzzling conclusion that the raw association between education and wages actually substantially *understates* the causal effect of education on wages. As Card (2001) documents, this conclusion about the returns to schooling is actually fairly common across diverse studies that use a variety of IV strategies and data sources. One explanation proposed by Card (2001) is that this arises from a failure to account for heterogeneity in the causal effect of schooling on wages.

We can use the `ivcrc` module to assess this explanation. The syntax is similar to that for the IV estimator:

```
. ivcrc wage76 (grade76 = col4 age76 agesq76) 'ControlVars', ///
>      dendog(exp76 expsq76)
(default settings do not compute standard errors, see bootstrap() option)
(estimating the conditional rank of grade76)
(estimating rule-of-thumb bandwidth)
(estimating beta(r) at each r[i] rank in the sample)

```

```
IVCRC                                     Number of obs      =      3,010
-----+-----
      wage76 |      Coef.   Std. Err.      z    P>|z|     [95% Conf. Interval]
-----+-----
      grade76 |   .0852767           .           .           .           .           .
...

```

Note: Average coefficients over R = [0,1] rank subset; Bandwidth = .0370327

We treat potential experience, `exp76`, as a derived endogenous variable here because it is defined as a deterministic function of `grade76` and `age76`. Whereas the coefficient on `grade76` reported by the standard linear regression estimator implemented by `ivregress` will estimate a difficult-to-interpret quantity like (3), the coefficient on `grade76` produces an estimator of the average causal effect of a one year increase in `grade76`. The causal effect estimated here of 8.53 percent is substantially lower than the linear IV estimate of 13.33 percent. This supports Card's (2001) reasoning if, as he argues, the usual linear IV estimator places more weight on individuals with higher returns to schooling. The `ivcrc` estimate is also somewhat larger than the linear regression coefficient 0.0725.

We now demonstrate some of the options for `ivcrc` by evaluating the statistical significance and robustness of the above ATE estimate. First, we compute standard errors, which tends to be time-consuming due to the necessity of using the bootstrap. The syntax and results are:

				Replications	=	100

wage76	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
-----+-----						
grade76	.0780691	.0302556	2.58	0.010	.0187693	.137369
...						

Note: Average coefficients over $R = [0,1]$ rank subset; Bandwidth = .025

we find that the results are also not sensitive to the functional form of the kernel, in concordance with the usual folklore for nonparametric kernel regression.

One interesting way to explore both the robustness and potential explanations for our finding is to change the set $\mathcal{R}$ over which the average is being taken. By default, `ivcrc` averages over all estimated conditional ranks ($\hat{R}_i$) directly as in (6). Alternatively, if we are concerned about results being driven by outliers in the conditional distribution of education, we can specify $\mathcal{R}$ to be $[.05, .95]$. Trimming the distribution in this way, while maintaining the smaller bandwidth and more accurate rank estimation from above,

```
. ivcrc wage76 (grade76 = col4 age76 agesq76) 'ControlVars', ///
>   dendog(exp76 expsq76) bootstrap(reps(100) seed(5282020)) bandwidth(.025) ///
>   ranks(200) average(.05(0).95)
(running _ivcrc_estimator on estimation sample)
```

Bootstrap replications (100)

```
-----+--- 1 ----+--- 2 ----+--- 3 ----+--- 4 ----+--- 5
..... 50
..... 100
```

IVCRC	Number of obs	=	3,010
	Replications	=	100

wage76	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
-----+-----						
grade76	.0735619	.0345893	2.13	0.033	.0057681	.1413558
...						

Note: Average coefficients over $R = [.05, .95]$ rank subset; Bandwidth = .025

we obtain slightly lower estimated returns to education and a slightly larger standard error, but overall similar results to the estimates which used the full observed conditional distribution of

6 Conclusion

In this paper we have discussed the CRC model, which is a parsimonious IV model that explicitly incorporates heterogeneous treatment effects. The `ivcrc` module for Stata implements an estimator for the CRC model. The estimator can be used to carefully analyze heterogeneous treatment effects in a way that the usual linear estimator implemented by `ivregress` cannot. Because the estimator is based on averaging conditional linear regressions, it scales well and incorporates covariates easily, making it attractive from a practical perspective.

There are a few important limitations to the estimator that are worth mentioning both as caveats, and as directions for future research. First, the conditions described in Section 2.3 to construct the control function, R , from the instrument, Z , require the endogenous variable, X , to be continuously distributed. The extent to which these methods fail for discrete endogenous variables is an interesting theoretical question that may have important implications for practice. Second, the `ivcrc` module implements a rule-of-thumb bandwidth. There may exist other, superior methods for automated bandwidth selection. Third, defining, detecting, and correcting for weak instruments is an unexplored topic for the CRC model.